Discrete & Continuous Data

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Discrete & continuous data

What is discrete and continuous data?

- Numerical data (quantitative data) can be either **continuous** or **discrete**
- **Continuous data** can take **any numerical value** on a scale
  - e.g. height, length, weight, mass
  - For continuous data the measurements can become more and more accurate the more you 'zoom in'
- **Discrete data** can only take **particular numerical values** on a scale
  - Often these are integers
    - e.g. numbers of people or objects
  - But they don't *have* to be integers
    - e.g. shoe sizes, which include 'half sizes'

Examiner Tips and Tricks

Do you need some sort of scale to measure it? If so, it's likely to be continuous data.

Help me understand

Worked Example

State whether the data in the following examples is discrete or continuous.

(a) The weights of dogs participating in a dog show.

Answer:

Weight can take *any* value, so it is continuous

Continuous

(b) The lengths of leaves taken from a particular tree.

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Continuous

(c) The number of computers owned by each household in a particular city.

Answer:

Only integer (i.e. whole number) values are possible, so it is discrete

Discrete

(d) The time taken, to the nearest hour, to complete a task by each person out of a group of people.

Answer:

Although this is time, it wants it 'to the nearest hour', not measured continuously
Therefore only certain values are possible, so it is discrete

Discrete

(e) The time taken to swim 100 metres by each member of a swimming club.

Answer:

Here there are no constraints placed on the measurement of time, therefore it could take any value, so is continuous

Continuous

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